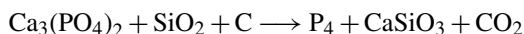


6.4 Problems

- Balance the following equations:
 - $\text{SO}_2 + \text{H}_2\text{O} \longrightarrow \text{H}_2\text{SO}_3$
 - $\text{H}_2\text{O} \longrightarrow \text{H}_2 + \text{O}_2$
 - $\text{PCl}_3 + \text{Cl}_2 \longrightarrow \text{PCl}_5$
 - $\text{C} + \text{O}_2 \longrightarrow \text{CO}$
 - $\text{Mg} + \text{O}_2 \longrightarrow \text{MgO}$
- Balance the following equations:
 - $\text{NH}_3 + \text{O}_2 \longrightarrow \text{N}_2 + \text{H}_2\text{O}$
 - $\text{I}_4\text{O}_9 \xrightarrow{\Delta} \text{I}_2\text{O}_5 + \text{I}_2 + \text{O}_2$
 - $\text{K}_2\text{S}_2\text{O}_3 + \text{Cl}_2 + \text{H}_2\text{O} \xrightarrow{\Delta} \text{KHSO}_4 + \text{HCl}$
 - $\text{CS}_2 + \text{O}_2 \longrightarrow \text{CO}_2 + \text{SO}_2$
- Write and balance equations for the following.
 - Calcium carbonate $\xrightarrow{\text{Heat}}$ Calcium oxide + Carbon dioxide
 - Potassium chlorate (KClO_3) $\xrightarrow{\Delta}$ Potassium chloride + Oxygen
 - Lithium nitride (Li_3N) $\xrightarrow{\Delta}$ Lithium + Nitrogen
 - Hydrogen peroxide $\xrightarrow{\Delta}$ Water + Oxygen
- Balance the following equations, which represent the-solid state synthesis of technologically important materials.
 - $\text{TiCl}_4 + \text{H}_2\text{S} \longrightarrow \text{TiS}_2 + \text{HCl}$
 - $\text{LaCl}_3 \cdot 7\text{H}_2\text{O} \longrightarrow \text{LaOCl} + \text{HCl} + \text{H}_2\text{O}$
 - $\text{CsCl} + \text{ScCl}_3 \longrightarrow \text{Cs}_3\text{Sc}_2\text{Cl}_9$
 - $\text{Li}_2\text{CO}_3 + \text{Fe}_2\text{O}_3 \longrightarrow \text{LiFe}_5\text{O}_8 + \text{CO}_2$
 - $\text{La}_2\text{O}_3 + \text{B}_2\text{O}_3 \longrightarrow \text{LaB}_6 + \text{O}_2$
 - $\text{ScCl}_3 + \text{SiO}_2 \rightleftharpoons \text{Sc}_2\text{Si}_2\text{O}_7 + \text{SiCl}_4$
- Balance the equations for the thermal decomposition of the following compounds:
 - $\text{NaN}_3 \longrightarrow \text{Na} + \text{N}_2$
 - $(\text{NH}_4)_2\text{Cr}_2\text{O}_7 \longrightarrow \text{Cr}_2\text{O}_3 + \text{N}_2 + \text{H}_2\text{O}$
 - $\text{Ag}_2\text{CO}_3 \longrightarrow \text{Ag} + \text{CO}_2 + \text{O}_2$
 - $\text{NaHCO}_3 \longrightarrow \text{Na}_2\text{CO}_3 + \text{CO}_2 + \text{H}_2\text{O}$
 - $\text{Al}_2(\text{CO}_3)_3 \longrightarrow \text{Al}_2\text{O}_3 + \text{CO}_2$
- Complete and balance the following neutralization reactions.
 - $\text{H}_2\text{SO}_4 + \text{KOH} \longrightarrow$
 - $\text{H}_3\text{PO}_4 + \text{Ba}(\text{OH})_2 \longrightarrow$
 - $\text{HBr} + \text{Sn}(\text{OH})_4 \longrightarrow$
 - $\text{HBr} + \text{NaOH} \longrightarrow$
 - $\text{H}_4\text{P}_2\text{O}_7 + \text{NaOH} \longrightarrow$
- Complete and balance the following double decomposition reactions.
 - $\text{Ca}(\text{NO}_3)_2 + \text{Na}_2\text{SO}_4 \longrightarrow$
 - $\text{Mg}(\text{NO}_3)_2 + \text{Li}_2\text{S} \longrightarrow$
 - $\text{Pb}(\text{NO}_3)_2 + \text{Cs}_2\text{CrO}_4 \longrightarrow$

- (d) $\text{FeCl}_3 + \text{CaS} \longrightarrow$
- (e) $\text{Na}_3\text{PO}_4 + \text{Zn}(\text{OH})_2 \longrightarrow$

8. The industrial production of phosphorus is represented by the following reaction:



Balance the equation for the reaction.

9. Hydrocarbons and their oxygen derivatives react with oxygen to produce carbon dioxide and water. The relative amounts of carbon dioxide and water formed depend on the composition of the parent hydrocarbon. Complete and balance the following equations:
- (a) $\text{C}_4\text{H}_8 + \text{O}_2 \longrightarrow$
 - (b) $\text{C}_6\text{H}_6 + \text{O}_2 \longrightarrow$
 - (c) $\text{C}_6\text{H}_{12}\text{O} + \text{O}_2 \longrightarrow$
 - (d) $\text{C}_{12}\text{H}_{22}\text{O}_{11} + \text{O}_2 \longrightarrow$
 - (e) $\text{C}_4\text{H}_{10} + \text{O}_2 \longrightarrow$
10. Write balanced chemical equations for the following reactions:
- (a) Iron and sulfur react at elevated temperature to form iron (III) sulfide.
 - (b) Octane, C_8H_{18} , a component of gasoline, burns in oxygen to produce carbon dioxide and water.
 - (c) Sulfur dioxide burns in oxygen to form sulfur trioxide.
 - (d) Barium hydroxide reacts with phosphoric acid to produce barium phosphate and water.
 - (e) Solid aluminum dissolves in hydrochloric acid to form aluminum chloride and hydrogen gas.
11. Write balanced chemical reactions for the following:
- (a) Aluminum reacts with chlorine to form aluminum chloride.
 - (b) Potassium reacts with water to form potassium hydroxide and hydrogen gas.
 - (c) Boron trichloride gas reacts with steam (gaseous H_2O) to form boron trihydroxide and hydrogen chloride.
 - (d) Silver nitrate solution reacts with sodium chloride to give aqueous sodium nitrate and precipitate of silver chloride.
 - (e) Silicon tetrafluoride reacts with sodium hydroxide to form sodium silicate (Na_4SiO_4), sodium fluoride, and water.
 - (f) Calcium hydroxide reacts with ammonium chloride to form ammonia, calcium chloride, and water.
12. Balance the following chemical reactions.
- (a) $\text{Si}_4\text{H}_{10} + \text{O}_2 \longrightarrow \text{SiO}_2 + \text{H}_2\text{O}$
 - (b) $\text{CH}_3\text{NO}_2 + \text{Cl}_2 \longrightarrow \text{CCl}_3\text{NO}_2 + \text{HCl}$
 - (c) $\text{NaF} + \text{CaO} + \text{H}_2\text{O} \longrightarrow \text{CaF}_2 + \text{NaOH}$
 - (d) $\text{Al}_4\text{C}_3 + \text{H}_2\text{O} \longrightarrow \text{Al}(\text{OH})_3 + \text{CH}_4$
 - (e) $\text{TiO}_2 + \text{B}_4\text{C} + \text{C} \longrightarrow \text{TiB}_2 + \text{CO}$
 - (f) $\text{C}_7\text{H}_{16}\text{O}_4\text{S}_2 + \text{O}_2 \longrightarrow \text{CO}_2 + \text{H}_2\text{O} + \text{SO}_2$